

# Mikołaj Mieszko Charchuta

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**Bioinformatician turning exploratory genomics into reproducible, production-grade pipelines** (Nextflow, Snakemake, Docker, Conda), specialising in chromosome conformation capture (Meta Hi-C, 3C-seq) and machine learning on bacterial pangenomes. Bridging academia and industry: Meta Hi-C for phage-therapy R&D at Phagos, an independently grant-funded project on the 3D genome of *B. thuringiensis*, and prior R&D in personalised medicine and cellular agriculture. Three publications and invited teaching on reproducibility; EU remote-first, open to B2B contracts and industrial PhD (CIFRE) partnerships.

## EDUCATION

### BOKU, VIENNA
MSc Bioinformatics (Erasmus)
Sep 2026 -- Jul 2027
Academic exchange; thesis project with Prof. Christian Stauffer.

### AMU, POZNAN
MSc in Bioinformatics
Oct 2025 -- Jul 2027
Thesis: *In silico analysis of Cry9 toxins and their genetic determinants in the Bacillus thuringiensis pangenome* (first-author article in preparation).

### AMU, POZNAN
BSc in Bioinformatics & Biotechnology
Oct 2021 -- Sep 2025
Thesis: *naïve methods for taxonomic classification from marker genes* — released as the NaïveR R package.

### RESEARCH EXCHANGES
Erasmus+ & National
2022 -- 2024
• **Charles Univ. / Czech Acad. Sci.**: meta-omics & stable isotope probing; CryoEM of cultivated-meat proteins.
• **Univ. of Gdansk & GUMed**: bioprotective SynComs; phage genomics.

## SKILLS

### REPRODUCIBILITY
Nextflow • Snakemake • Docker
Conda • Git • CI/CD • Slurm • AWS

### PROGRAMMING
R • Python • Bash • C++
R package development (NaïveR)

### BIOINFORMATICS
3D genomics (Hi-C, Meta Hi-C, 3C-seq)
Pangenomics & gene mining (samtools, GATK)
Machine learning (scikit-learn, tidymodels)
Transcriptomics (DESeq2) • metagenomics (QIIME2, mothur, NaïveR)

### GRANTS
Study@Research, AMU (2026–27)
SPUN intl. grant (2022–24): ML for soil biodiversity hotspots (field video)

### LANGUAGES
English (C2) • Polish (Native)
German (B2) • French (B2)
Czech (A2) • Hungarian (A2)

## TALKS & COMMUNITY

### CRABS 2026 / HEALTH INTEL.
Invited Speaker & Workshop Lead
Aug 2026 | Kyiv, UA
Lecture on *Reproducibility in Computational Genomics*; 2 Nextflow pipeline workshops; talk on Meta Hi-C & ML for bacterial genomes.

### BIOINF. COMMUNITY PL
Co-organiser
2024 -- Present
Event coordination, website, open-science advocacy.

## EXPERIENCE

### PHAGOS | Bioinformatician
Jul 2026 -- Present | Remote / Paris, FR
- Deliver phage–host association calls into the therapeutic discovery pipeline by analysing Meta Hi-C data end-to-end, from raw reads to ranked pairings.
- Made analyses re-runnable and free of environment drift by rebuilding ad-hoc scripts as containerised Nextflow workflows (Docker, Conda) used across the R&D team.
- Guide two research projects and joint analyses with external academic partners (incl. Institut Pasteur), aligning experimental design with computational feasibility.

### AMU, DEPT. OF MICROBIOLOGY | Contract Researcher
Sep 2025 -- Jan 2028 | Remote / Poznań, PL
- Won competitive Study@Research funding (app. nr. 201/34/UAM/0074) to run an independent project on the 3D genome of *B. thuringiensis*, owning Hi-C library prep, sequencing and analysis solo.
- Produced the dataset behind a first-author manuscript by mining the *B. thuringiensis* pangenome for Cry9 toxin determinants.
- Standardised the group's Hi-C/3C-seq analysis with reusable containerised workflows for contact-map QC and scaffolding.

### DANUBE INT. SCHOOL VIENNA | Laboratory Tech / Science Mentor
Jun 2026 -- Aug 2028 | Vienna, AT
- Teach science through practical experiment: design and run hands-on labs, and mentor students through independent research projects.

### GESELLSCHAFT FÜR BIOANALYTIK | Scientist
Jul 2024 -- Jun 2026 | Poznań, PL
- Delivered allergen and biosafety analyses to commercial turnaround and QA standards, from field, through lab to computational.

### BENE MEAT TECHNOLOGIES | Bioinformatician
Mar 2024 -- Jul 2024 | Prague, CZ
- Supported cultivated-meat cell line R&D with transcriptomics, nanopore analyses and bioreactor automation.

### LEIDEN UNIVERSITY MEDICAL CENTER | Bioinformatician
Jun 2023 -- Oct 2023 | Leiden, NL
- Contributed to two peer-reviewed publications by building the pipelines and figures for a murine MASLD study. Conducted animal research.

### MNM DIAGNOSTICS | Genome Analyst
Feb 2022 -- Jul 2022 | Poznań, PL
- Interpreted clinical whole-genome data for personalised oncology, reporting variants to clinicians.

## PUBLICATIONS

**In silico analysis of Cry9 toxins and their genetic determinants in the *Bacillus thuringiensis* pangenome.** **Charchuta, M. M.** (first author), Baranek, J., et al. *In preparation, 2026.*

**Combination of dietary fiber and exercise training improves fat loss in mice but does not ameliorate MASLD more than exercise alone.** Kovynev, A.; **Charchuta, M. M.**; et al. *Am. J. Physiol. Gastrointest. Liver Physiol.*, 2025.

**WED-517 Oatmeal a day does not keep the doctor away: inulin and exercise impact on MASLD amelioration and gut-liver crosstalk.** Kovynev, A.; **Charchuta, M.**; et al. *J. Hepatol.*, 2024.

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